# Krishen Mathur

Champaign, IL | krishen2@illinois.edu | +1 (447) 902-6709 | linkedin.com/in/krishenmathur | github.com/krishenmathur | krishenmathur1000.github.io

## Education

**University of Illinois at Urbana-Champaign** — Expected December 2027

*B.S. Electrical Engineering, Minor in Semiconductor Engineering GPA: 3.50* — *Champaign, IL*

- **Relevant Coursework:** Analog Signal Processing, Digital/Analog Circuits, Semiconductor Electronics, Computer-Aided Design (PCB), Computer Systems and Programming, Probability and Statistics

## Projects

**Analog DJ Mixing Board** — May 2026 – Present

*Personal Project | Analog Circuit Design, PCB Design*

- Designing a 2-channel analog DJ mixer with op-amp gain stages, summing amplifier, 3-band EQ, and crossfader modules, progressing from breadboard prototype to fabricated PCB.
- Laying out a multi-layer PCB in **EagleCAD** with attention to ground planes and power decoupling to minimize channel crosstalk and noise coupling.
- Characterizing each module on the bench with oscilloscope, function generator, and DMM, measuring frequency response, gain, THD, and noise floor against SPICE-simulated targets.
- Built a **Python** workflow to process measurement data, generate Bode plots, and automate comparison of measured vs. simulated response.

**Project Stealth – Wearable EMG & IMU System** — Aug 2025 – Present

*I-MADE (Illinois Medical Advancements through Design and Engineering)* — *Champaign, IL*

- Engineered a wearable EMG + IMU data acquisition system to capture muscle fatigue and movement signals, integrating analog front-end design, Arduino microcontroller programming, and BLE data transmission.
- Designed a custom signal-conditioning circuit (instrumentation amplifier, active filters) to amplify and filter EMG signals for high-fidelity biosignal capture.
- Developed a real-time **Python** pipeline (serial ingestion, filtering, plotting) for live signal visualization; demoed to 300+ attendees at Engineering Open House and presented at the Cozad New Venture Challenge.
- Wrote embedded firmware to synchronize multi-channel EMG/IMU sampling, optimize power consumption, and stream data to CSV for offline analysis.

## Work Experience

**Power Electronics Intern – Sourcing & Development** — Summer 2026

*Centre for Innovation, Technology and Learning (CITL) – Hero MotoCorp* — *Jaipur, India*

- Simulated a custom active PFC boost converter in LTspice for a 580W Vida onboard charger, analyzing two-loop average current mode control
- Studied full onboard charger architecture across all modules (EMI, rectification, PFC, LLC resonant, control)
- Authored a concept proposal for MagSafe-style wireless charging for EVs
- Developed EBOMs and toured SMT/assembly lines to study end-to-end OBC production and vendor negotiations

**Team Lead, Student Technology Consultants** — Mar 2025 – Present

*University of Illinois at Urbana-Champaign – Technology Services* — *Champaign, IL*

- Promoted to Team Lead after one year as a Student Technology Consultant; oversee a team of consultants and coordinate coverage and escalations to keep daily support operations running smoothly.
- Train new hires on diagnostic workflows, ticketing procedures, and university systems, accelerating ramp-up and improving first-contact resolution quality.
- Handle complex, escalated tickets spanning networking, account access, software, and hardware issues that require deeper troubleshooting than frontline triage.
- Manage AV and institutional technology systems across campus spaces, supporting classroom, event, and enterprise technology deployments.

## Technical Skills

**Hardware & Circuits:** Analog Circuit Design, PCB Design (EagleCAD), Op-Amp Circuits, Signal Conditioning, BMS, Soldering, Breadboarding
**Measurement & Test:** Oscilloscope, Function Generator, DMM, Frequency Response Characterization (Bode, THD, Noise Floor), SPICE Simulation, Hardware Debugging
**Programming:** Python (data analysis, signal processing), C, C++, Assembly, Arduino, Embedded Firmware, Serial Communication, BLE
**Software & Tools:** Git/GitHub, Vivado, Autodesk Fusion 360, LaTeX, Linux
**Other:** Communication, Public Speaking, Billiards, Table Tennis